Single-Cell RNA-seq Analysis of Breast Cancer

Cellular heterogeneity and exploratory classification of healthy mammary tissue and ER-positive tumors

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Abstract

Single-cell RNA sequencing (scRNA-seq) provides a means of examining cellular diversity and condition-associated expression patterns at cellular resolution. This project analyzes six samples from the public GSE161529 breast-tissue dataset: three healthy mammary tissue samples and three estrogen receptor-positive (ER+) breast tumors. Following quality control, 36,490 cells were retained for normalization, feature selection, dimensionality reduction, batch-aware neighborhood construction, clustering, cell-type annotation, and differential-expression analysis. Seventeen transcriptional clusters were annotated. A set of 244 candidate genes across eight eligible cell populations was then represented as 1,952 sample-level, cell-type-specific mean-expression features. Logistic regression and Random Forest were compared using leave-one-sample-out cross-validation. Random Forest achieved an accuracy of 0.833 and an F1 score of 0.800, compared with 0.667 and 0.500 for logistic regression. These results illustrate an integrated workflow connecting cellular characterization with interpretable sample-level modeling. Given the small cohort and feature selection performed before cross-validation, classification performance should be considered exploratory.

1 Introduction

Breast tissue contains diverse epithelial, stromal, endothelial, and immune populations whose transcriptional states may change in disease. Single-cell RNA sequencing enables these populations to be examined separately, providing a more detailed view of tissue composition than bulk expression measurements. Published single-cell atlases of human breast tissue provide an important context for investigating this heterogeneity [1].

This project applies a Python-based scRNA-seq workflow to a subset of the public GSE161529 dataset [2]. The objectives were to characterize transcriptional clusters in healthy mammary tissue and ER+ breast tumors, annotate cell populations using marker-gene evidence, identify condition-associated expression differences within selected populations, and explore sample-level classification using cell-type-specific expression features.

2 Dataset and quality control

Six biological samples were analyzed: three healthy mammary tissue samples and three ER+ tumor samples. The merged count matrix contained 39,101 cells and 33,538 gene features before cell-level filtering. Quality control used total counts, detected genes, and mitochondrial-read percentage. Sample-specific thresholds were applied to total counts and detected-gene counts, with an upper detected-gene threshold to reduce likely multiplets and a global mitochondrial cutoff of 15%.

Table 1: Samples included in the analysis and cells retained after quality control.

GEO accession	Condition	Retained cells
GSM4909253	Healthy	4,812
GSM4909254	Healthy	6,482
GSM4909265	Healthy	5,368
GSM4909296	ER+ tumor	7,119
GSM4909297	ER+ tumor	7,103
GSM4909298	ER+ tumor	5,606
Total		**36,490**

Figures 1 and 2 show the distributions of the principal quality-control metrics before and after filtering. The retained cells had a median mitochondrial-read percentage of 3.17%, with all cells at or below the 15% cutoff.

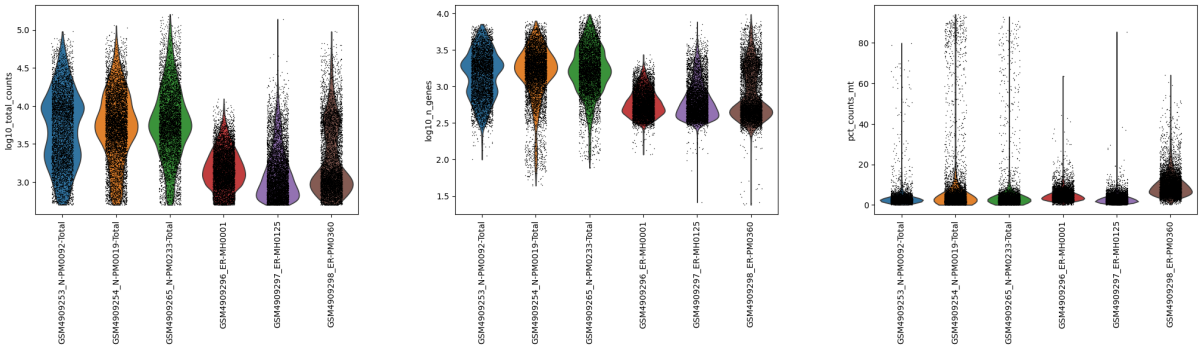


Figure 1: Per-sample distributions of quality-control metrics before filtering.

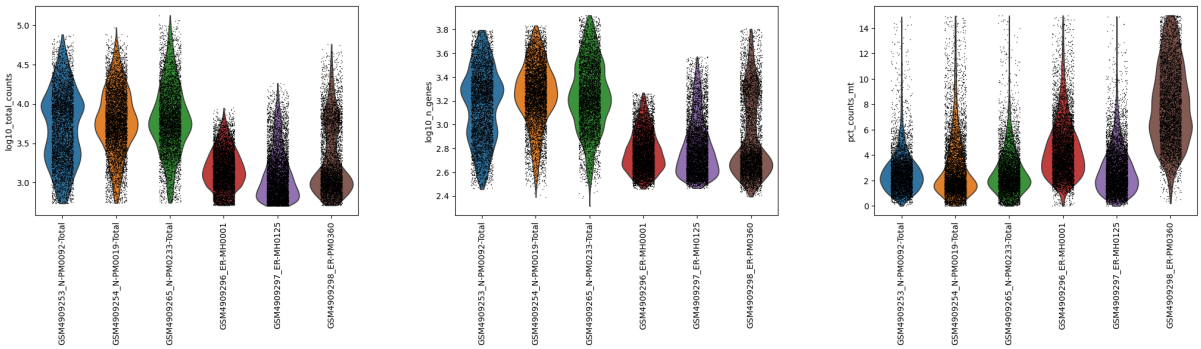


Figure 2: Quality-control metric distributions after filtering.

3 Methods

3.1 Preprocessing and dimensionality reduction

Counts were retained in a dedicated layer, normalized to 10,000 counts per cell, and log-transformed. The Seurat v3 method was used to select 2,000 highly variable genes, incorporating sample identity in feature selection. The selected expression matrix was scaled and used for principal component analysis (PCA). Scanpy provided the core data-processing and visualization workflow [3].

3.2 Batch-aware clustering and annotation

A batch-balanced neighborhood graph was constructed with BBKNN, using sample identity as the batch variable [4]. UMAP was used to visualize the neighborhood structure, and Leiden clustering at resolution 0.8 identified 17 transcriptional clusters [5, 6]. Cluster markers were ranked using Wilcoxon testing and interpreted alongside canonical marker-gene expression to assign descriptive cell-type labels.

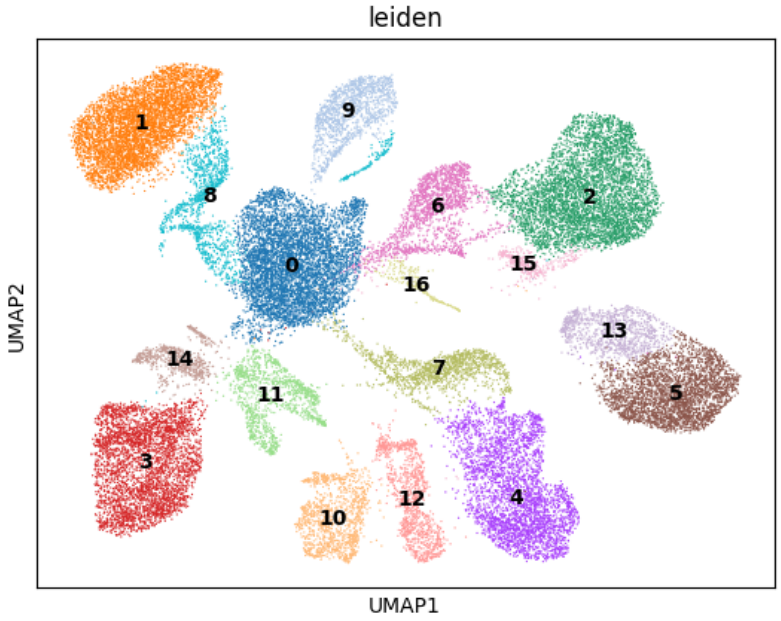


Figure 3: UMAP visualization of the 17 Leiden clusters identified using the BBKNN neighborhood graph.

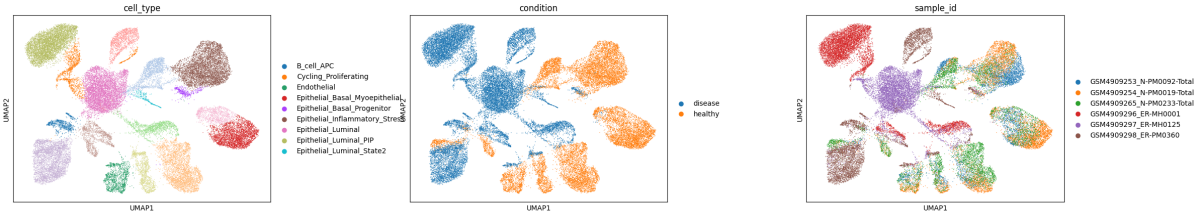


Figure 4: UMAP embeddings showing annotated cell populations, condition, and sample identity.

3.3 Differential expression and feature construction

Within-cell-type comparisons were performed between healthy and ER+ tumor cells for populations containing at least 50 cells from each condition. Differential expression was assessed using the Wilcoxon rank-sum test. Candidate genes were selected using Benjamini–Hochberg-adjusted $p < 0.05$ and an absolute log fold change of at least 0.5 [7]. Eight cell populations met the cell-count criterion. Selecting up to 50 genes per eligible population yielded 244 unique genes.

For each sample and eligible cell population, mean log-normalized expression was calculated for the selected genes. This produced 1,952 cell-type-specific features per sample, retaining the cellular context of each gene rather than combining all expression values into a single profile.

3.4 Classification and evaluation

Two classifiers were compared: L2-regularized logistic regression and Random Forest. Performance was evaluated using leave-one-sample-out cross-validation, holding out one biological sample in each fold and training on the other five. Accuracy, precision, recall, F1 score, ROC-AUC, and confusion matrices were used to summarize the held-out predictions.

4 Results

4.1 Cellular structure and annotation

Quality control retained 36,490 cells across the six samples. The 2,000 highly variable genes provided the feature space for PCA and neighborhood analysis. BBKNN-based graph construction followed by Leiden clustering identified 17 transcriptional clusters. Marker-informed annotation assigned descriptive labels spanning epithelial states, fibroblast and stromal populations, endothelial cells, immune-associated populations, and a cycling population. The annotated UMAPs show how these populations are distributed across conditions and samples (Figures 3 and 4).

4.2 Differential expression and classification features

Within-cell-type differential-expression analysis was restricted to eight populations with sufficient cell counts in both conditions. Applying the adjusted-p-value and effect-size thresholds yielded 244 unique candidate genes. Representing these genes separately across the eight eligible populations produced 1,952 sample-level features.

4.3 Sample-level classification

Table 2 summarizes the model results. Random Forest achieved higher accuracy, recall, and F1 score than logistic regression. Both models had a reported ROC-AUC of 1.000; given the small number of samples, this value should be interpreted cautiously.

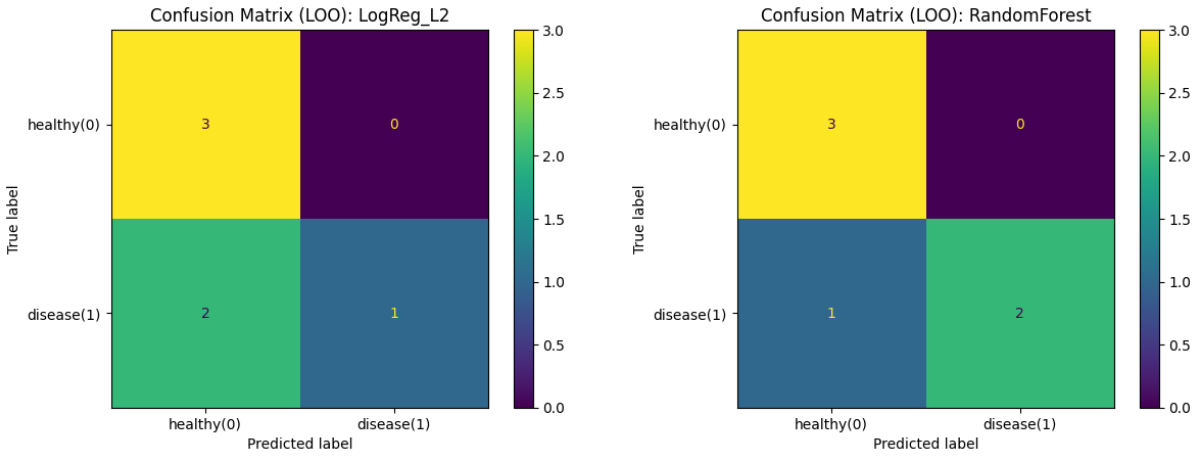
Table 2: Leave-one-sample-out classification metrics.

Model	Accuracy	Precision	Recall	F1 score	ROC-AUC
Logistic regression	0.667	1.000	0.333	0.500	1.000
Random Forest	0.833	1.000	0.667	0.800	1.000

5 Discussion

The workflow links cell-level characterization with sample-level modeling. Clustering and marker-informed annotation describe a range of epithelial, stromal, endothelial, and immune-associated populations. The feature-construction strategy retains both gene identity and cell-type context, making the resulting predictors easier to interpret biologically.

Random Forest showed stronger threshold-based performance in this evaluation, correctly classifying five of the six samples. Logistic regression provided directly interpretable coefficients, which can be related to cell-type-specific expression patterns. Ranked features included signals



(a) Logistic regression

(b) Random Forest

Figure 5: Confusion matrices aggregated over the six held-out samples. Logistic regression produced 3 true negatives, 0 false positives, 2 false negatives, and 1 true positive. Random Forest produced 3 true negatives, 0 false positives, 1 false negative, and 2 true positives.

from stromal, immune-associated, and epithelial populations. These observations are consistent with the broader importance of the tumor microenvironment in breast tissue, but should be treated as candidate associations rather than evidence of causal mechanisms.

The classification results remain exploratory for two reasons. First, six samples provide very limited evidence about generalization to new patients. Second, differential-expression-based feature selection was performed before cross-validation, meaning that the held-out samples influenced the selected feature set. A more independent evaluation would repeat feature selection within each training fold and assess the complete pipeline in additional samples. Differential-expression analyses of single-cell data should also account for biological replication when making population-level inferences [8].

6 Conclusion

This project integrated scRNA-seq quality control, feature selection, dimensionality reduction, batch-aware clustering, cell-type annotation, differential expression, and exploratory classification. The analysis retained 36,490 cells and identified 17 transcriptional clusters. A cell-type-specific feature representation based on 244 genes was used to compare logistic regression and Random Forest at the sample level. Random Forest achieved the stronger accuracy and F1 score in the reported evaluation. The principal value of the workflow is its integration of cellular characterization with interpretable expression features, providing a foundation for more rigorous validation on larger cohorts.

Data availability

The source dataset is available through the NCBI Gene Expression Omnibus under accession GSE161529. Raw data and large intermediate objects are not included with this report.

References

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